The Economics of Brexit

Lecture 2: Customs unions

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Macroeconomic Applications and Policy

Context

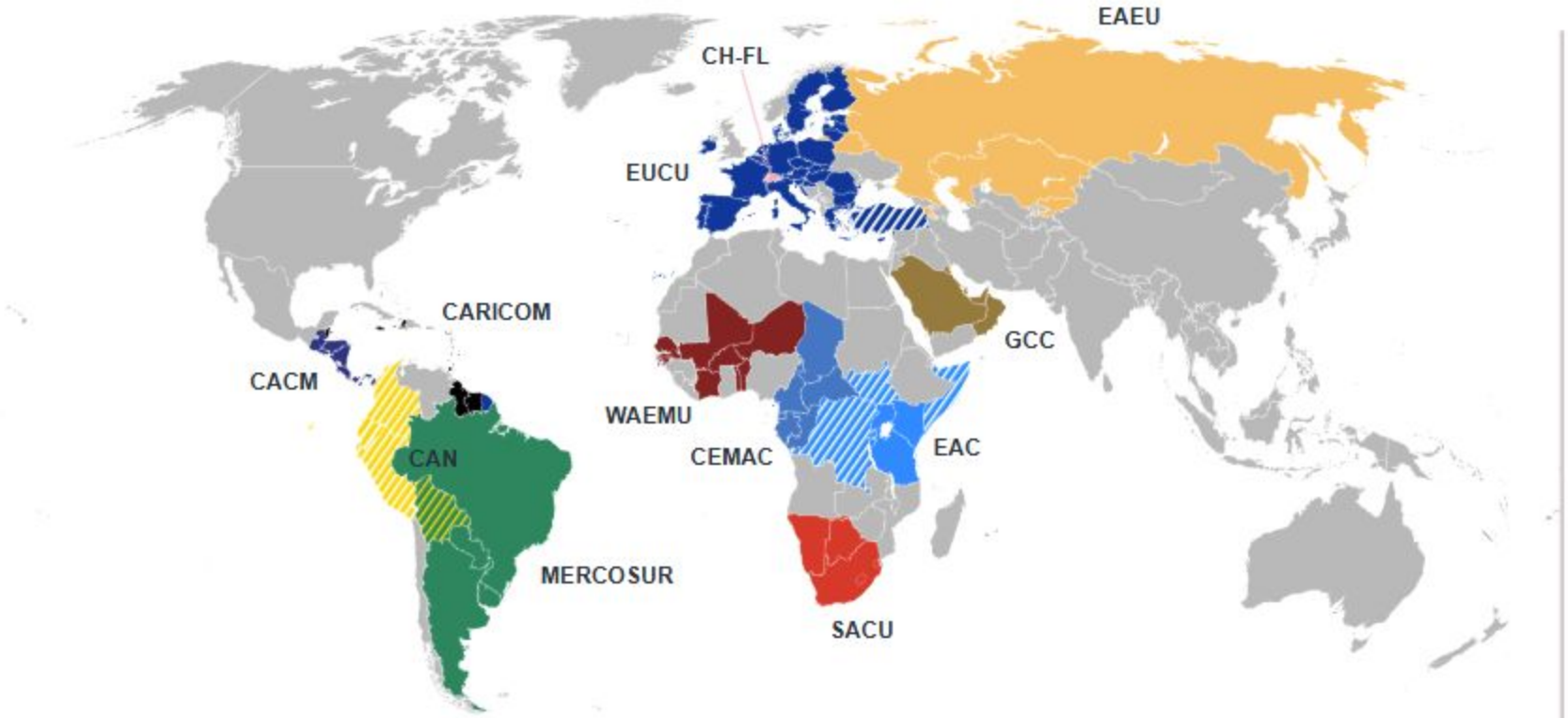
- One of the main goals of the Treaty of Rome was "the **elimination**, as between Member States, **of customs duties** and quantitative restrictions on the imports and export of goods, **and of all other measures having equivalent effect**" and "the establishment of a **common customs tariff and of a common commercial policy towards third countries**"

Learning objectives

After this lecture, you will be able to:

1. Analyse the welfare effects of joining a customs union:
   - Removal of **tariff barriers** and establishment of a common external tariff (1958–1968).
   - Removal of **non-tariff barriers** (1986–1992).
2. Understand the economic incentives to join a preferential trade agreement.
3. Discuss evidence on the welfare impact of customs unions.

Customs unions



Explore a trade bloc

Hover over a coloured region, or select its abbreviation below.

A customs union removes internal tariffs and adopts a common external tariff.

Current membership: October 2026.

EUCU | EAEU | MERCOSUR | CAN | CACM | CARICOM | WAEMU | CEMAC | EAC | SACU | GCC | CH-FL

Checked 6 October 2026 · Selected arrangements; implementation varies. Stripes: special status or overlapping membership.
Hover for details and official sources. Map base (adapted; territories simplified).

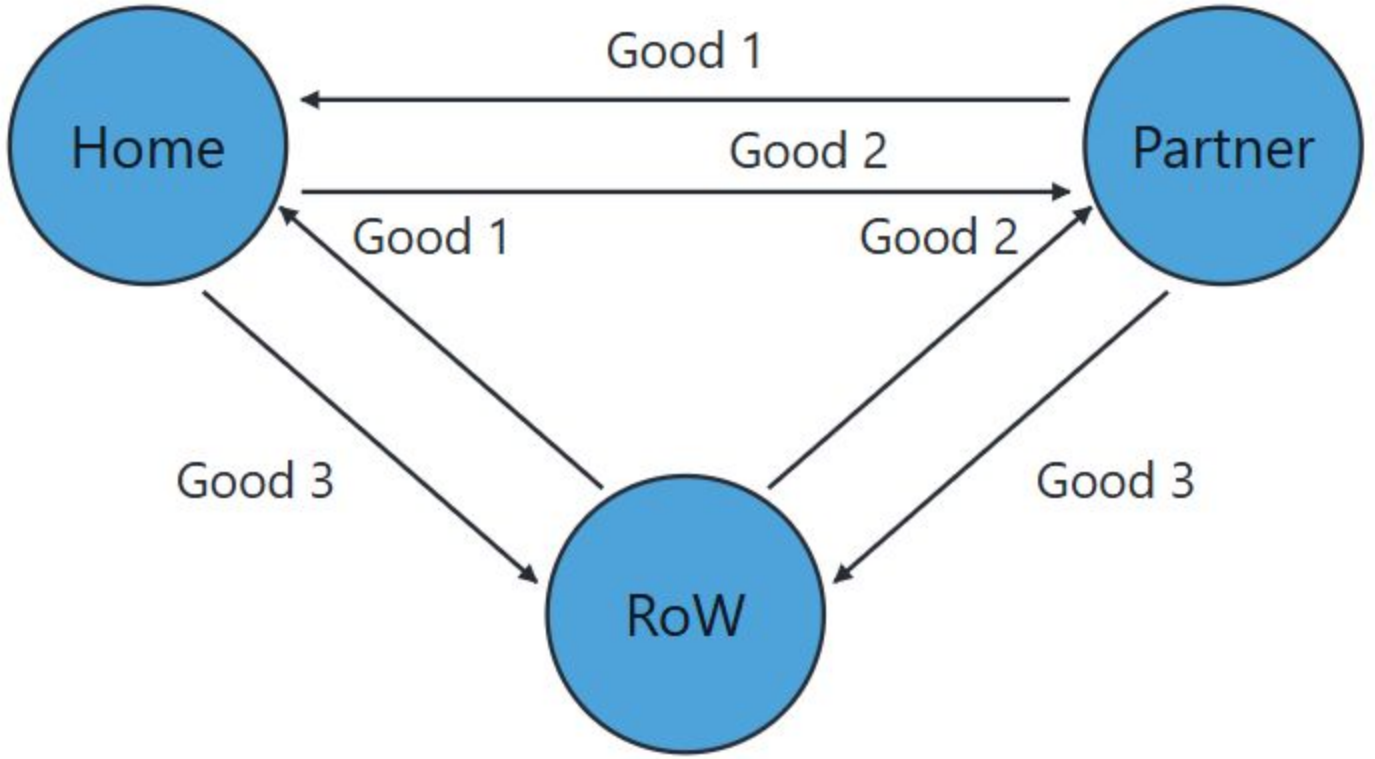
Does removing tariff barriers benefit every country?

Framework

- Perfect competition
- Full employment
- No transport costs
- Perfect factor mobility (capital and labour)
- Firms optimise (P=MR=MC)
- Consumers optimise (P=MU)

Customs union

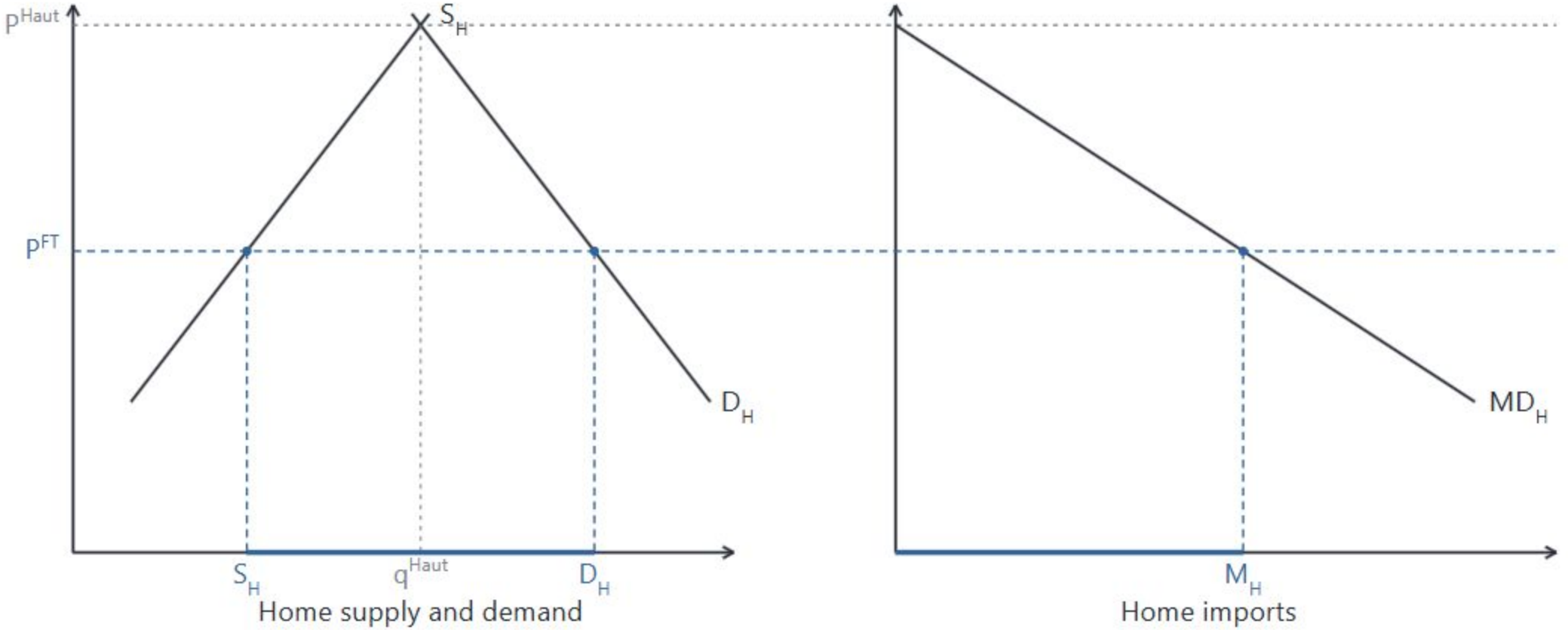
A simple, symmetric framework



Home imports good 1; Partner imports good 2; RoW imports good 3.

The import demand curve

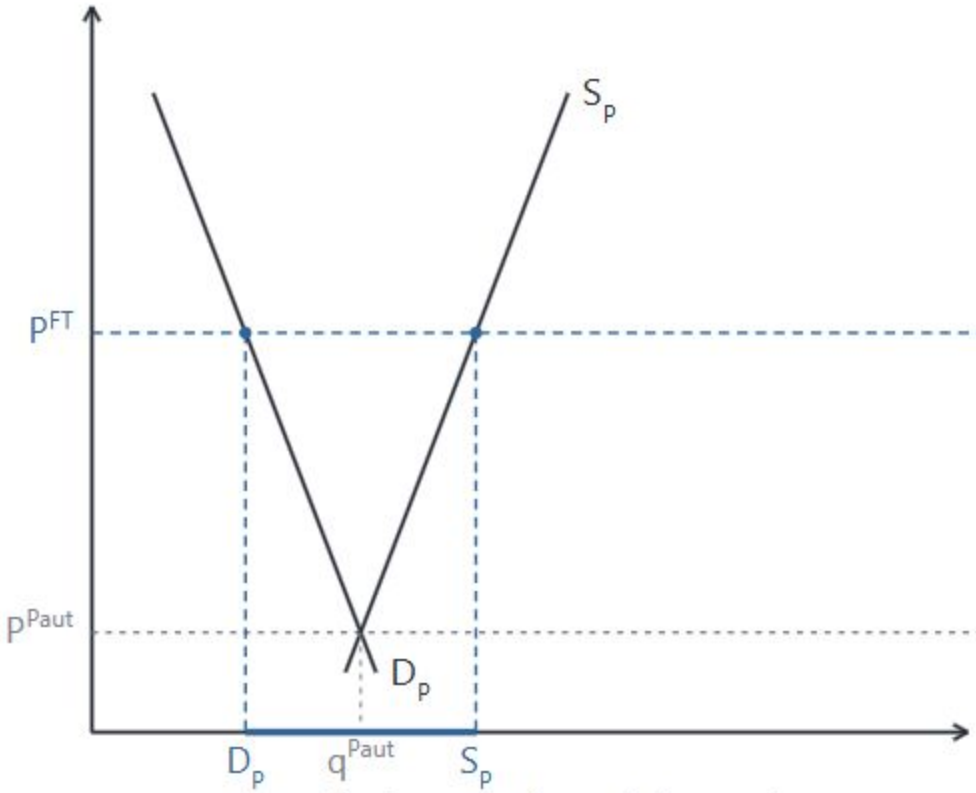
Below Home's autarky price, imports equal domestic demand minus domestic supply: $M_H \equiv D_H - S_H$.



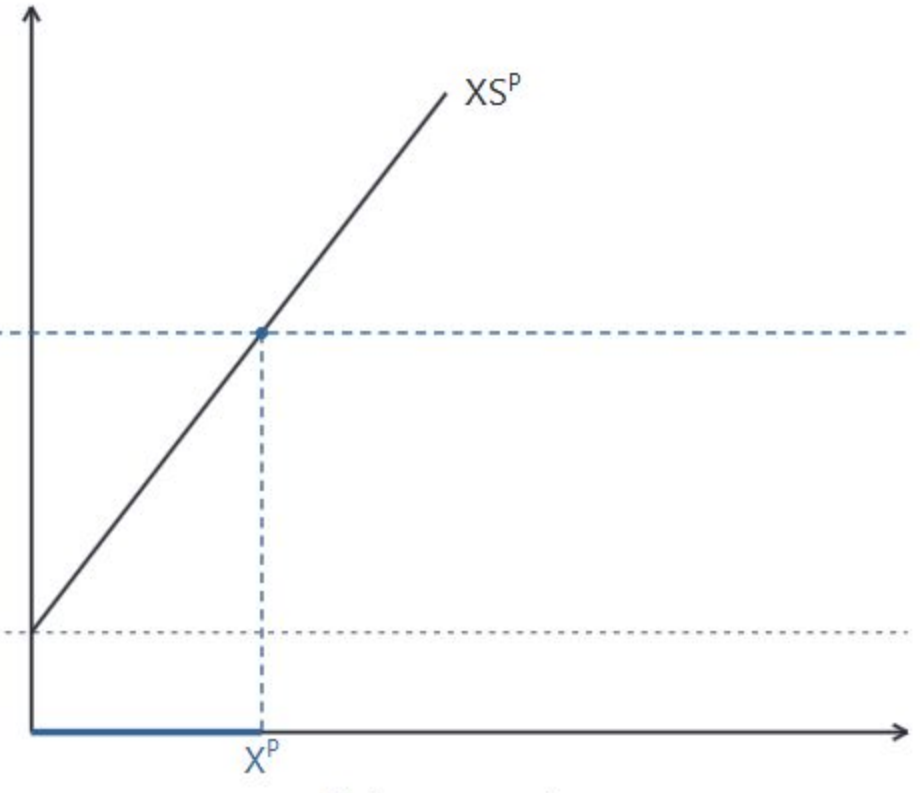
The highlighted quantity intervals are equal: **imports are the gap between demand and supply**.

Partner's export supply curve

Above Partner's autarky price, exports equal domestic supply minus domestic demand: $X_P \equiv S_P - D_P$.



Partner supply and demand

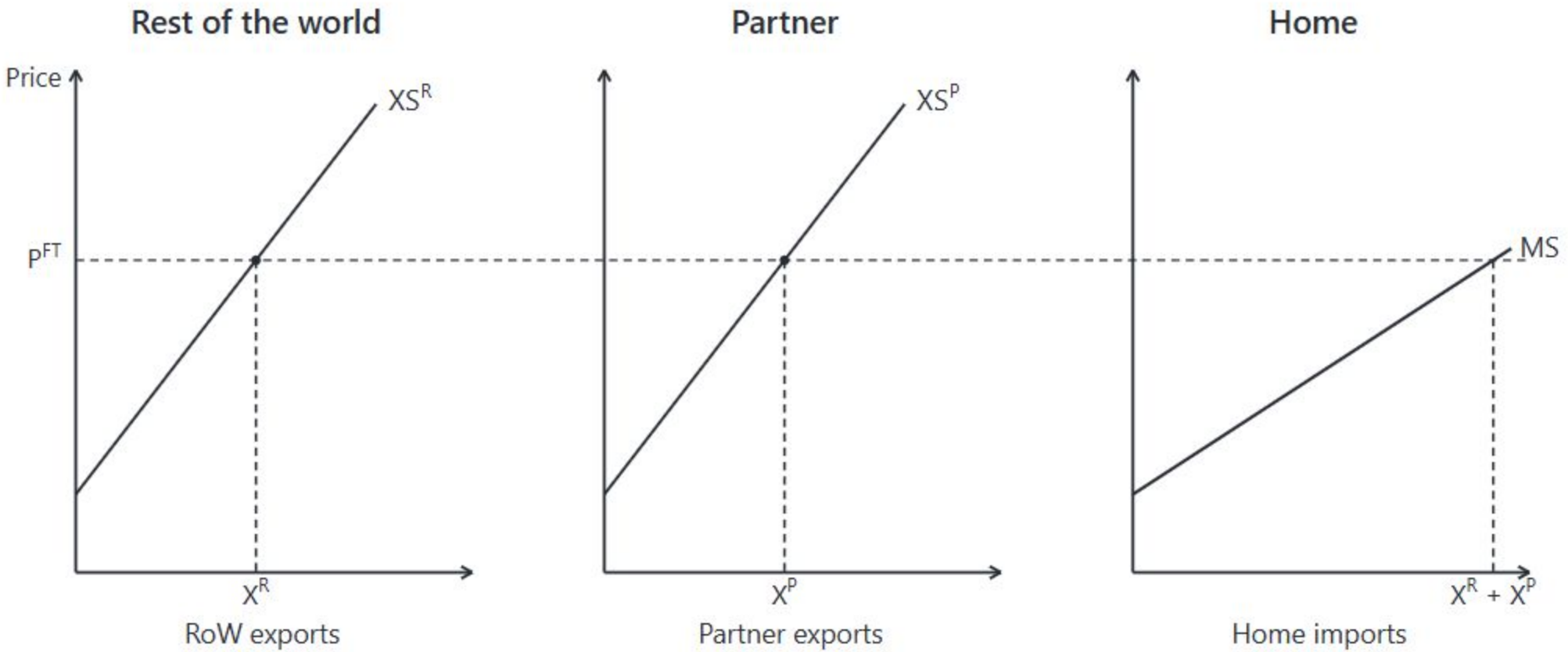


Partner exports

The highlighted quantity intervals are equal: **exports are the gap between supply and demand**.

The aggregate export supply curve

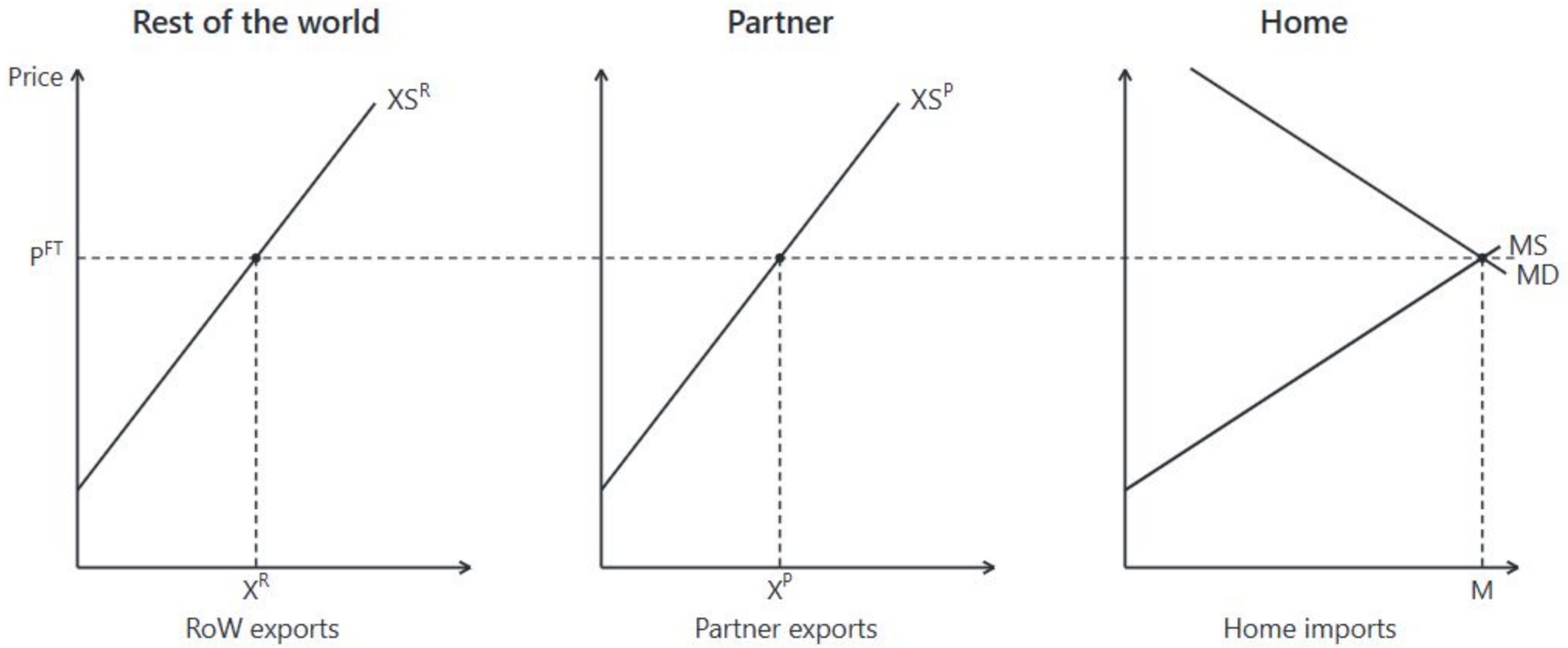
Home's import supply is the horizontal sum of both export supply curves: $MS \equiv XS^R + XS^P$.



At each common price, **add quantities horizontally** to obtain aggregate supply.

World equilibrium (free trade)

At the free-trade equilibrium price P^{FT}, world demand equals world supply.



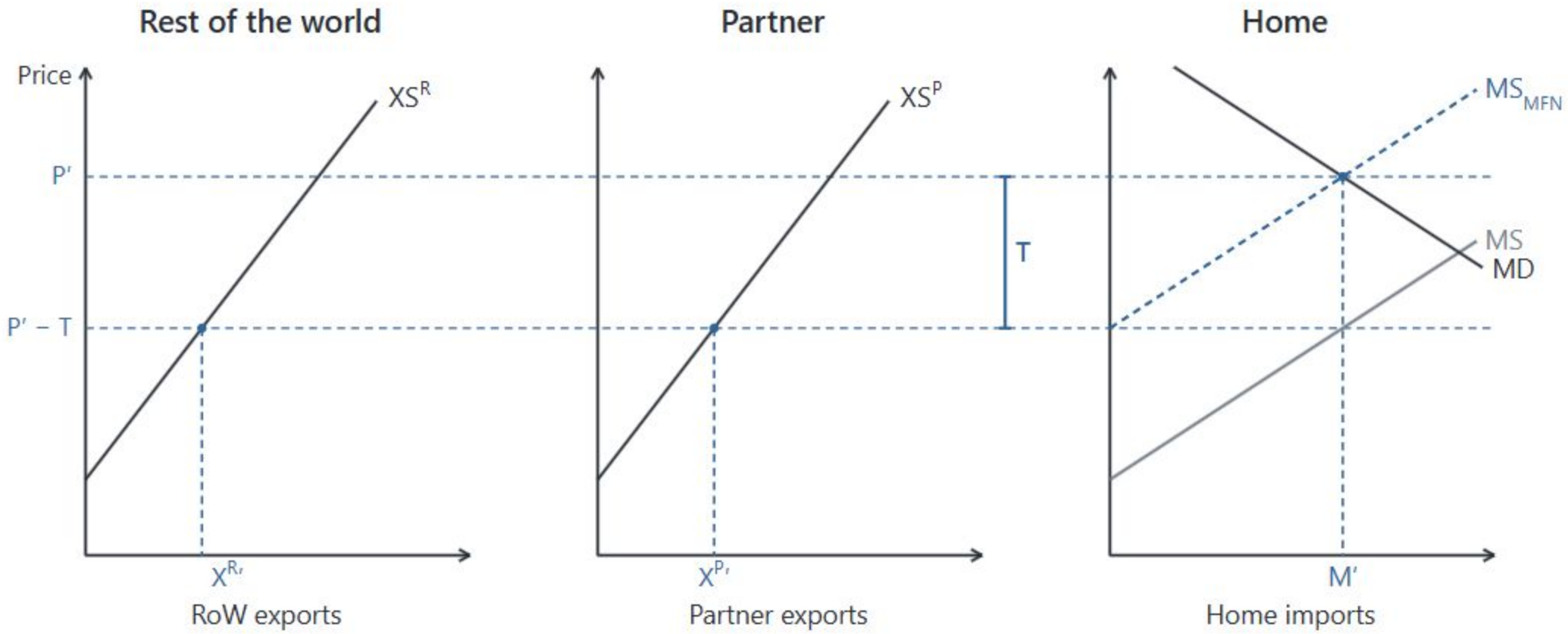
$$D_H + D_P + D_R = S_H + S_P + S_R \quad \Leftrightarrow \quad M_H = X_P + X_R.$$

Before the formation of the CU

- Assume that Home, Partner and RoW were all applying the 'Most Favoured Nation' principle (GATT/WTO), according to which trade barriers should not be discriminatory.
- Home imposes the same tariff on Partner and RoW.
- The "welfare" impact of the imposition of a MFN tariff is negative from a global point of view: the imposing country might gain, but the affected country always loses (see Baldwin and Wyplosz).

Three-country analysis (MFN)

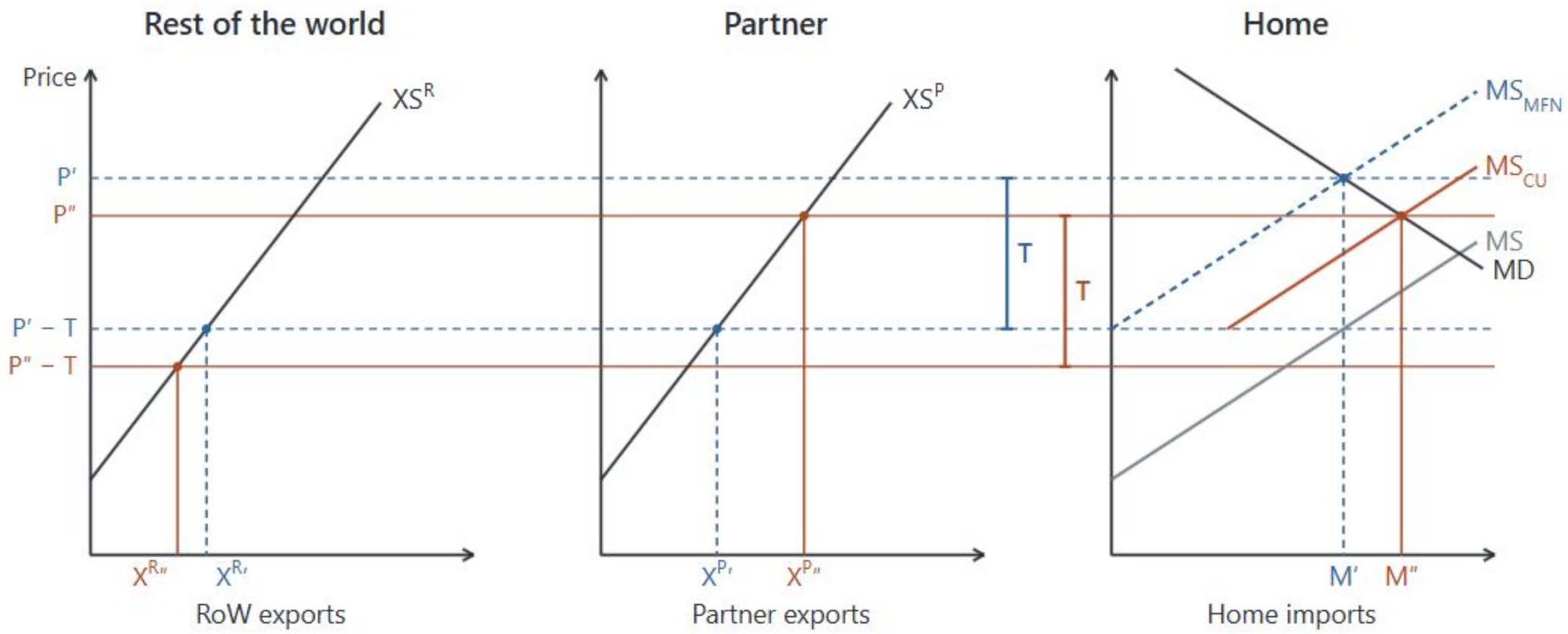
Home applies the same tariff T to Partner and the rest of the world.



Predict: what changes when Home removes the tariff on Partner?

The formation of the CU

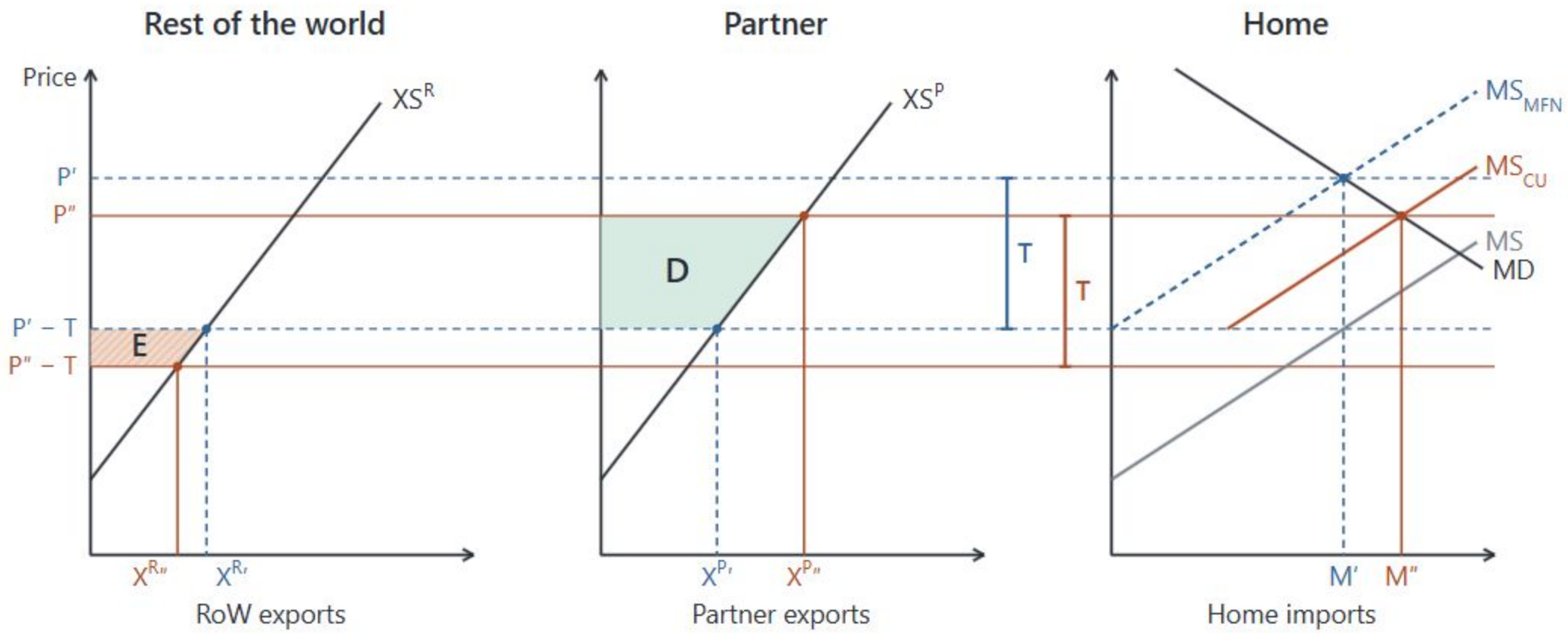
Home removes the tariff on Partner. The tariff on RoW remains T.



Home's price falls; Partner exports more; RoW exports less. **Dashed blue: before. Solid orange: after.**

“Welfare” effects

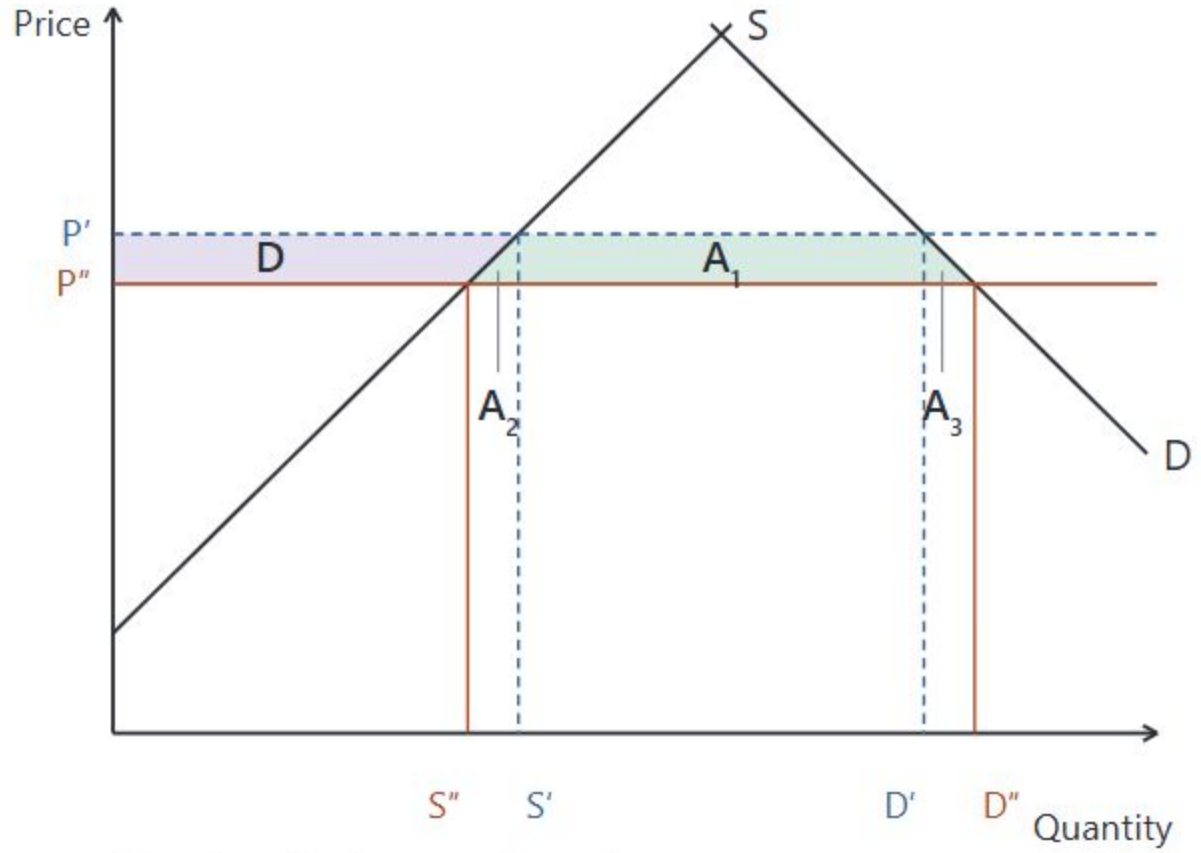
The shaded areas measure each exporter’s net welfare change for this good.



Partner gains D. RoW loses E. Home’s welfare needs a separate calculation.

Home welfare: consumers and producers

Home's price falls from P' to P'': consumers gain and producers lose.



D is a transfer from producers to consumers.

Consumer surplus

$+D + A_2 + A_1 + A_3$

Producer surplus

$-D$

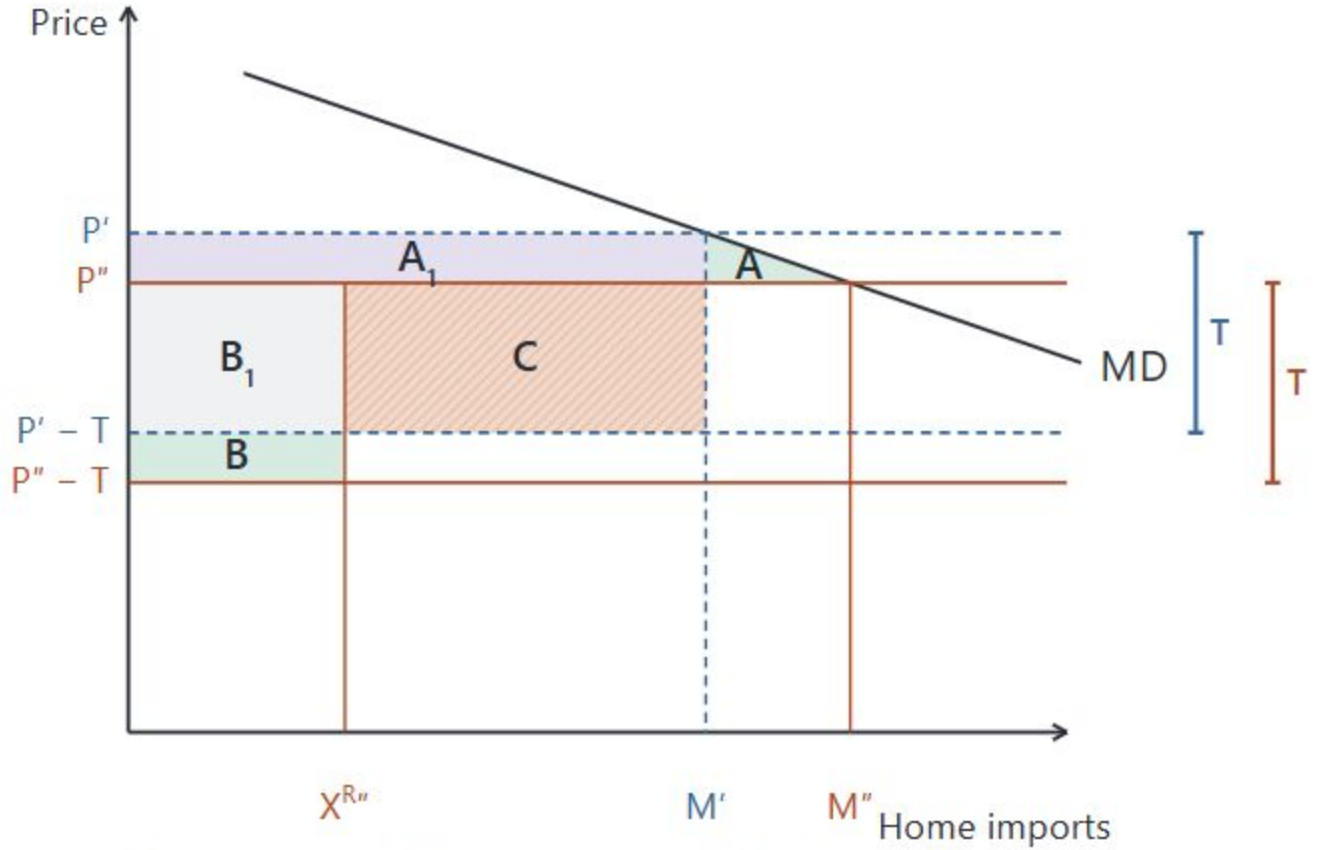
Net private surplus

$A_2 + A_1 + A_3$

The transfer D cancels. We still need the change in **tariff revenue**.

Home welfare: tariff revenue and total

After the customs union, only imports from RoW generate tariff revenue.



Green: gains A + B. Hatched orange: loss C.

Tariff revenue

Before: $A_1 + B_1 + C = TM'$.

After: $B_1 + B = TX^{R''}$.

Change: $B - A_1 - C$.

Add private surplus

$$A_2 + A_1 + A_3 + B - A_1 - C$$
$$= A_2 + A_3 + B - C$$
$$= A + B - C.$$

Home gains if $A + B > C$.
Home loses if $A + B < C$.

Summing up the welfare effects

Home

- Good 1 (imported): $A + B - C$.
- Good 2 (exported to Partner): $+D$.
- Good 3 (exported to RoW): no change.

Partner

Identical to Home.

Rest of the world

- Goods 1 and 2 (exported to customs union members): $-2E$.
- Good 3 (imported from customs union members): no change.

Summing up the welfare effects: in words

Three elements

- **Adam Smith's certitude**: foreign firms gain when tariffs against them are eliminated (increase in the price of the exported good);
- **Haberler's spillover**: third nations – those excluded from the tariff reductions – lose (fall in the price of the exported good);
- **Viner's ambiguity**: "trade creation" effect with Partner has to be weighed against "supply switching" or "trade diversion".

What do we mean by welfare?

Non-tariff barriers

Protection by country

Country	Tariff rate		Non-tariff barrier rate	
	Rate[1]	Standard deviation	Rate[1]	Standard deviation
Belgium	7.0	7.6	19.6	28.2
Denmark	7.1	4.1	18.2	27.0
Germany, West	7.4	6.0	22.3	27.4
Greece	7.0	8.6	25.5	25.8
Spain	6.8	4.8	13.9	22.1
France	7.4	14.3	18.4	26.1
Ireland	7.5	7.5	20.8	27.0
Italy	7.6	14.4	20.9	27.1
Netherlands	7.1	7.6	20.6	28.1
Portugal	7.1	3.4	19.1	20.2

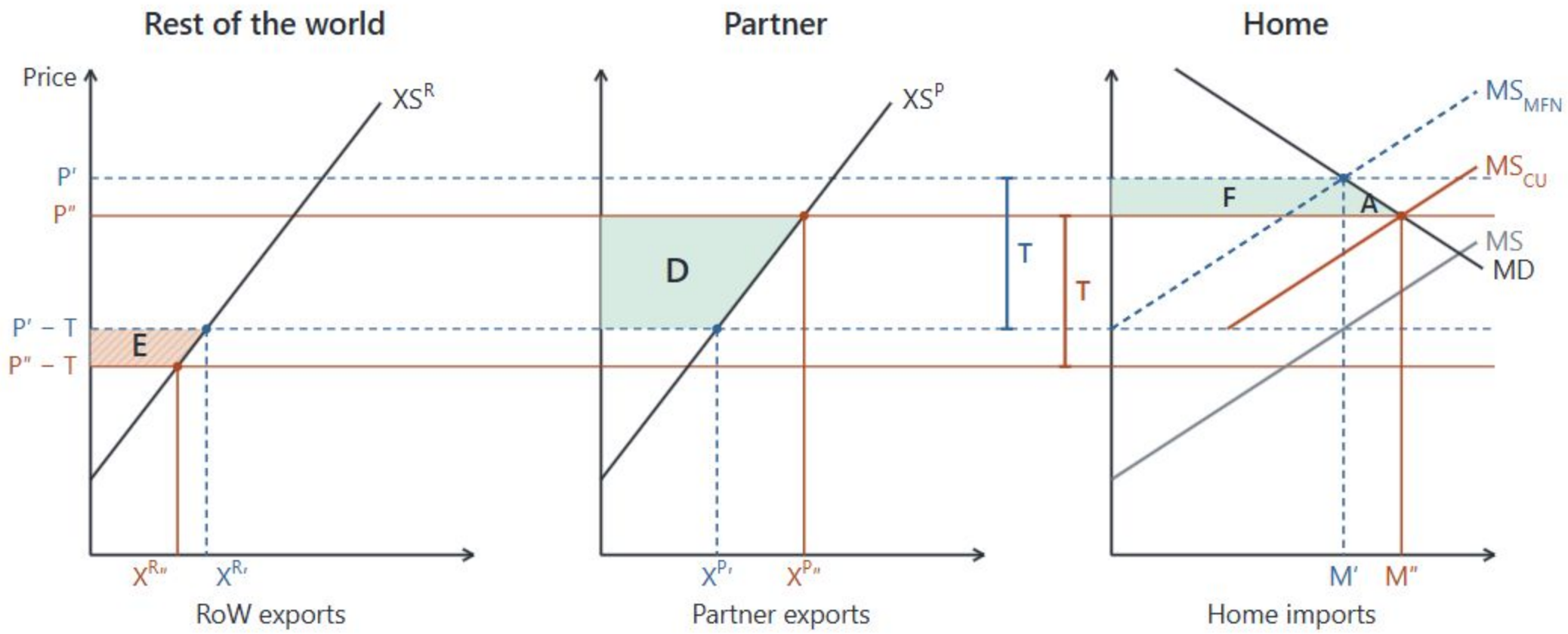
Protection by industry

Industry	Tariff rate		Non-tariff barrier rate	
	Rate[1]	Standard deviation	Rate[1]	Standard deviation
Food products	9.8	33.6	45.9	30.0
Textiles	11.7	24.5	69.8	38.2
Apparel	12.3	30.4	71.7	35.7
Footwear	13.3	44.5	33.8	41.9
Furniture	6.4	46.2	0.9	46.5
Industrial chemicals	10.2	23.5	9.1	30.1
Iron & steel	9.8	38.0	47.7	34.6
Machinery, electric	8.6	28.3	14.3	33.3
Transport equipment	7.9	30.7	25.5	38.8
Professional & scientific equipment	6.5	23.6	2.7	30.3

[1] Import-weighted measure. Source: Lee and Swagel (1997), as reproduced in the original lecture. Historical figures.

Non-tariff barriers: welfare effects

Here T is a real per-unit trade cost. Home removes it on Partner's goods; it remains on RoW goods.



Home gains $A + F$; Partner gains D; RoW loses E. A real trade cost generates no tariff revenue.

The incentives to join a CU

We have seen that removing non-tariff barriers produces unambiguously positive welfare effects in this framework. This is not the case for tariff barriers.

Question

Why join a customs union? Would **unilateral liberalisation** towards the potential Partner be better?

Answer

Not necessarily: a country would not benefit from Partner's tariff reduction within the customs union (**area D**).

What did the early studies find?

> **Trade creation exceeded trade diversion in Europe, but the estimated gains were small.**
>
> Baldwin and Venables (1995).

Findings in context

- Little diversion: much of Europe's trade was already intra-European.
- Limited creation: initial trade barriers were low.
- The perfect-competition framework has limitations.

Empirical challenges

- **Ex ante:** quantifying effects in a theoretical framework is complicated (e.g. Miller et al., 1977).
- **Ex post:** attributing increased intra-European trade, or reduced RoW trade, to customs-union membership is difficult (e.g. Winters, 1987).

Next lecture

- We will talk about the competition effects of the Single Market Act.
- Tip: revise the theory of the firm under monopoly and oligopoly (Cournot competition).

References

Essential reading

- Baldwin and Wyplosz, *The Economics of European Integration*, **Chapters 4 and 5 (any edition)**.

Further reading

See the ECON2271 Talis reading list, also available through Blackboard.